Interpretable Fine-Grained Aircraft Classification Network for Remote Sensing Image With Image Pair Interaction and Neural Tree

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***Abstract*—This article proposes a novel interpretable framework for fine-grained aircraft classification in high-stakes remote sensing applications. Our approach addresses three key challenges: small interclass variance, large intraclass variance, and the need for model interpretability. Specifically, our framework is built on the Swin transformer (SwinT) backbone and includes three main modules. First, we present the dynamic attention fusion module (DAFM), which adaptively fuses multistage attention maps from the SwinT backbone. By leveraging a dispersion-based weighting mechanism, DAFM balances the contributions of coarse and fine-grained features, capturing both global structures and localized details. Second, we propose the adaptive image pair interaction (AIPI) module, which dynamically adjusts feature interaction strategies based on intraclass and interclass similarity, effectively enhancing informative regions and improving robustness. To further optimize discriminative power, we incorporate an AIPI loss function that enforces intraclass consistency and interclass separability. Finally, we develop a binary neural tree module (BNTM) to hierarchically select and propagate informative image patches, enhancing both feature refinement and interpretability through explicit path-based decision-making. Extensive experiments on benchmark datasets demonstrate that our framework significantly improves classification accuracy and interpretability, making it well-suited for applications requiring transparent and reliable decision-making. The code can be found at https://github.com/StarmanGzx/BNTM**

***Index Terms*—Attention map fusion, fine-grained aircraft classification, interpretability, neural tree, remote sensing, transparency in decision-making, vision transformer (ViT).**

I. Introduction

With the continuous advancement of remote sensing satellite technology and the increasing sophistication of modern warfare, the task of fine-grained aircraft classification in remote sensing imagery has become critical in both civilian and military domains [1], [2], [3], [4], [5], [6], [7], [8], [9], [10].



Fig. 1. Illustration of fine-grained classification in comparison with the general classification.

Unlike general object recognition tasks, which typically involve both detection and classification, fine-grained classification focuses solely on categorizing targets into subcategories [11], [12], [13]. It operates on precropped images that contain only the target object (e.g., an aircraft), without involving any object detection procedures. As illustrated in Fig. 1, compared to general classification (e.g., distinguishing between airplanes, ships, and vehicles), fine-grained classification aims to differentiate visually similar subcategories within the same major class, such as various aircraft models. This makes the task considerably more challenging due to subtle interclass differences and potentially large intraclass variations [14], [15], [16].

In the field of fine-grained aircraft classification, early studies primarily relied on handcrafted feature extraction methods, such as scale invariant feature transform (SIFT) [17] and histogram of oriented gradient (HOG) [18]. While these methods were capable of capturing local features of aircraft to some extent, they showed limited performance when dealing with viewpoint variations, scale differences, and complex backgrounds in remote sensing images.

In recent years, with the rise of deep learning, approaches based on a convolutional neural network (CNN) have been widely applied for image feature extraction and high-precision classification [19], [20], [21], [22], [23]. These models effectively capture complex visual patterns through hier-

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archical feature learning, overcoming the limitations of traditional methods in handling image details. For example, Lin et al. [24] proposed bilinear CNN, which enhanced fine-grained classification by fusing features from different levels. Wu et al. [25] proposed a deep attention-based CNN model with bilinear pooling and spatial long short term memorys (LSTMs) for fine-grained classification. Zuo et al. [26] proposed a deep attention-based CNN model with bilinear pooling and spatial LSTMs for fine-grained classification. Fan et al. [27] proposed an unsupervised method with dual-level progressive similar instance selection for better feature learning. FGA-YOLO [28] improved fine-grained aircraft classification through multiscale fusion and dynamic sample balancing. Qian et al. [29] enhanced fine-grained aircraft classification by adaptively inferring prior knowledge and optimizing CNN feature extraction.

However, because CNN primarily rely on local receptive fields for feature extraction, they may focus on smaller and discontinuous regions of the image, resulting in insufficient capture of critical distinguishing features, such as subtle differences in aircraft models. Therefore, CNN-based methods still face limitations when addressing issues such as large intraclass variation and small interclass differences in remote sensing images.

Recently, transformer-based models have achieved breakthroughs in computer vision tasks [30]. Vision transformer (ViT), such as the one proposed by Dosovitskiy [31], leverages self-attention mechanisms to model long-range dependencies between image patches, providing new insights for fine-grained classification. Variants like the Swin transformer (SwinT) [32] further enhanced computational efficiency and feature extraction capabilities by introducing a local window mechanism. Wang et al. [33] introduced a ViT-based pyramid architecture that enhances remote sensing object detection and classification through spatial-reduction attention. He et al. [34] proposed TransFG, a transformer-based model for fine-grained classification, using attention maps and contrastive loss to select discriminative patches. Yu et al. [35] proposed a mixing attentive ViT to improve fine-grained visual classification (FGVC) performance. Zhang et al. [36] proposed a ViT-based model with patch combination, region filter, and token integration for FGVC.

Although compared to CNN-based methods, ViT-based methods capture global context and maintain a broader receptive field through self-attention, enabling them to effectively recognize complex visual patterns [31]. However, they require a large amount of diverse training data to fully exploit their self-attention capabilities. Furthermore, the challenges of large intraclass variation and small interclass differences require deeper exploration of the relationships between images [37]. Unfortunately, both CNN- and ViT-based methods lack the intrinsic ability to effectively model these relationships, which is crucial for addressing the subtle variations present within aircraft categories.

In addition, both ViT-based and CNN-based methods lack interpretability, which is crucial in key domains such as environmental monitoring, military reconnaissance, and medical diagnostics [38]. Since it is currently impossible to develop a model that is 100% accurate, model transparency—defined as the ability for humans to observe the decision-making process and promptly identify potential errors—becomes paramount. This transparency determines whether human intervention is needed during decision-making, thereby enhancing the model's fault tolerance in high-risk tasks.

In conclusion, we have summarized the challenges of fine-grained aircraft classification in optical remote sensing imagery for practical applications, as illustrated in Fig. 2.

1) *Small Interclass Variances:* For example, the various classes of the Boeing series exhibit only minor differences in wing and fuselage shapes or sizes, as well as engine placements, resulting in minimal visual distinctions between categories, making them challenging to differentiate.
2) *Large Intraclass Variances:* Aircraft of the same type exhibit significant appearance variations under different viewpoints, scales, lighting conditions, and noise levels.
3) *Imbalanced Training Samples:* For example, in a specific real-world dataset, the category with the highest number of training samples has 9779 instances, while the category with the fewest has only 17.
4) *Black-Box Inference:* Most existing deep learning models operate as black-box, providing high classification accuracy but offering little insight into the decision-making process.

To overcome the challenges mentioned earlier, we present an enhanced framework built upon the SwinT backbone. This model incorporates innovative techniques aimed at addressing the limitations of existing methods in fine-grained classification. By integrating attention map fusion, facilitating interactions between image pairs, and enhancing the transparency of the decision-making process, our approach effectively mitigates issues such as small interclass variance, large intraclass variance, class imbalance, and lack of interpretability. The key contributions of this work are as follows.

1) We propose the dynamic attention fusion module (DAFM), which adaptively fuses attention maps based on their dispersion across different stages of ViT backbone, effectively balancing coarse and fine-grained information to enhance complex pattern classification in fine-grained classification.
2) We propose the adaptive image pair interaction (AIPI) module, which dynamically adjusts feature interactions based on intraclass and interclass similarity, enhancing informative regions to address small interclass variance and large intraclass variance. The corresponding AIPI loss promotes intraclass consistency and interclass separability, improving model robustness and discrimination.
3) We propose a novel binary neural tree module (BNTM) to enhance model interpretability. BNTM adopts a binary tree structure, where the most discriminative image region is progressively selected at each layer to form an explicit decision path from root to leaf. This path reveals the key regions the model attends to during inference, thereby making the decision process more transparent and the model more interpretable.

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4) We evaluate our approach on two benchmark datasets, FAIR1M-1.0 [39] and RarePlanes [40]. FAIR1M-1.0 is characterized by small interclass variance and large intraclass variance, while RarePlanes presents a highly imbalanced scenario. Experimental results demonstrate that our method outperforms state-of-the-art approaches, effectively addressing the challenges of fine-grained aircraft classification and providing interpretable visual analyses.

The remainder of this article is organized as follows. We describe the related works in Section II and introduce our proposed approach in Section III. Evaluations on two public datasets are presented in Section IV, followed by conclusion in Section V.

II. Related Works

A. Fine-Grained Aircraft Classification

Fine-grained aircraft classification has been a challenging task in the field of remote sensing due to the small interclass variance and large intraclass variance present in aircraft images. Traditional machine learning methods, such as support vector machines (SVMs) [41] and random forests [42], initially played a role in aircraft classification but struggled to handle the complex variations within the same aircraft class, such as different angles, scales, and lighting conditions.

Recent researches have shifted toward deep learning approaches, particularly CNN, which have shown significant improvements in extracting discriminative features from aircraft images. For example, Li et al. [43] introduced a bilinear channel fusion network for feature extraction, and an adaptive classification loss to optimize the fine-grained classification. Chen et al. [12] proposed a knowledge-driven deep learning method for aircraft classification, based on CNN-based segmentation and classification networks. Yi et al. [44] improved fine-grained classification with a CNN by using a miner for feature purification and a refiner for augmentation, achieving strong results. Nie et al. [45] proposed a classifier-adaptive Earth mover's distance with attention mechanisms for few-shot fine-grained aircraft classification, achieving state-of-the-art results. And Ouyang et al. [46] used prototypical contrastive learning to enhance fine-grained detection in remote sensing images. However, the inherent limitations of CNN-based methods, such as their inability to effectively capture global contextual information and their sensitivity to subtle variations, often result in suboptimal performance for fine-grained aircraft classification.

B. ViT in FGVC

ViT has emerged as a powerful alternative to CNN, offering significant advantages in fine-grained visual tasks by leveraging the self-attention mechanism. Unlike traditional CNN-based methods, which focus on local feature extraction through convolution, ViT-based methods capture global dependencies between image patches, making them particularly effective for tasks like fine-grained aircraft classification where subtle differences in global context are crucial. For example, Fu et al. [47] proposed a progressive learning ViT (PLViT) that utilizes a coarse-to-fine optimization strategy to enhance open set classification in fine-grained remote sensing object classification. Zhang et al. [48] proposed a ViT model that enhances fine-grained classification performance through attention patch combination, critical regions filtering, and complementary tokens integration. Cheng et al. [49] employed two transformer-based branches for fine-grained classification and oriented localization, enhancing feature refinement with a spatial and channel transformer. Wang et al. [50] proposed a hybrid CNN-ViT architecture, P2FEViT, to enhance fine-grained remote sensing image classification by integrating local and global feature representations. However, most existing ViT-based approaches lack the effective fusion of attention across different stages of the network. The absence of attention fusion mechanisms may lead to the model failing to fully capture both local and global contextual information simultaneously, reducing its ability to differentiate between aircraft with subtle interclass variations and large intraclass differences.

C. Interpretability in FGVC

Current deep learning models based on CNN and ViT are often considered black-box, with opaque decision-making processes [51]. In remote sensing, understanding model decisions is vital. Posthoc explainable techniques like class activation mapping (CAM) and Grad-CAM [52], [53], [54], [55] offer some interpretability, but only after training and inference. These methods do not provide real-time transparency during the decision-making process.

Subsequently, researchers introduced prototype-based explainable methods [56], [57], [58] to address the limitations of posthoc interpretability. These methods work by learning representative patches, or prototypes, for each class during the training phase. During inference, the model compares input images with these learned prototypes to make decisions. This approach offers a more transparent way of understanding how the model arrives at its conclusions, as every prediction can be traced back to the most relevant prototypes. For example, Guo et al. [38] proposed a multiscale rotation-invariant prototype network that enhances fine-grained aircraft classification accuracy and interpretability without requiring part annotations. However, a critical challenge with prototype-based methods is the need for a substantial number of samples to effectively learn the prototypes for each class, which poses difficulties in accurately capturing the representative features of classes with limited samples, resulting in low classification accuracy for those categories.

III. Methods

A. Overview

As illustrated in Fig. 3, the overall framework of our method consists of two parallel branches: one is the SwinT backbone integrated with the DAFM and AIPI modules, and the other is the BNTM designed for interpretability. Specifically, the SwinT backbone comprises four sequential stages, each containing a different number of SwinT blocks. The input image

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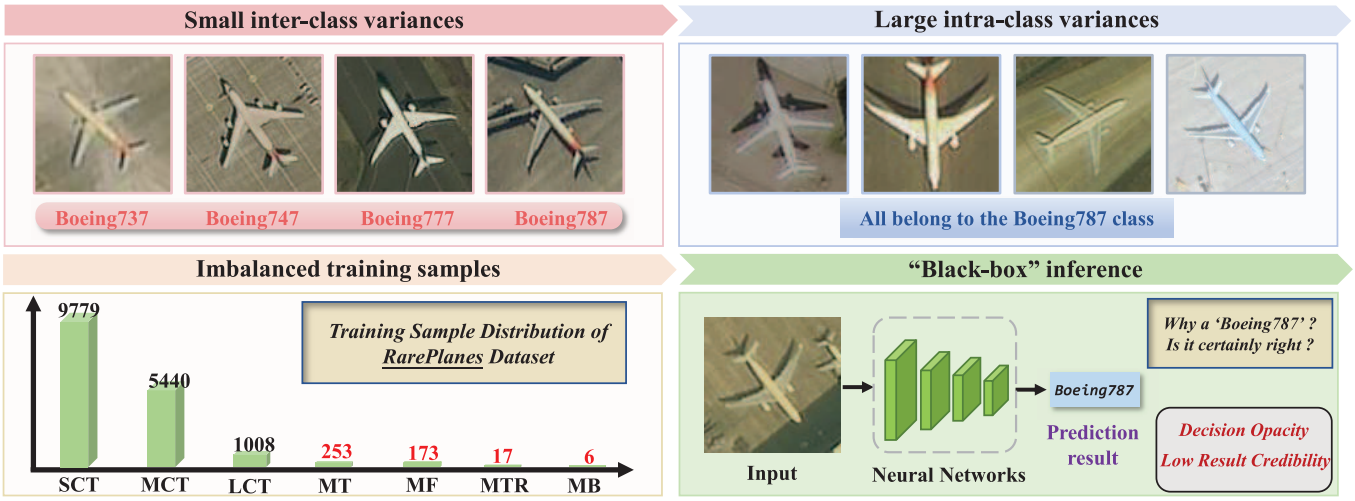


Fig. 2. Key challenges in fine-grained aircraft classification for remote sensing images.

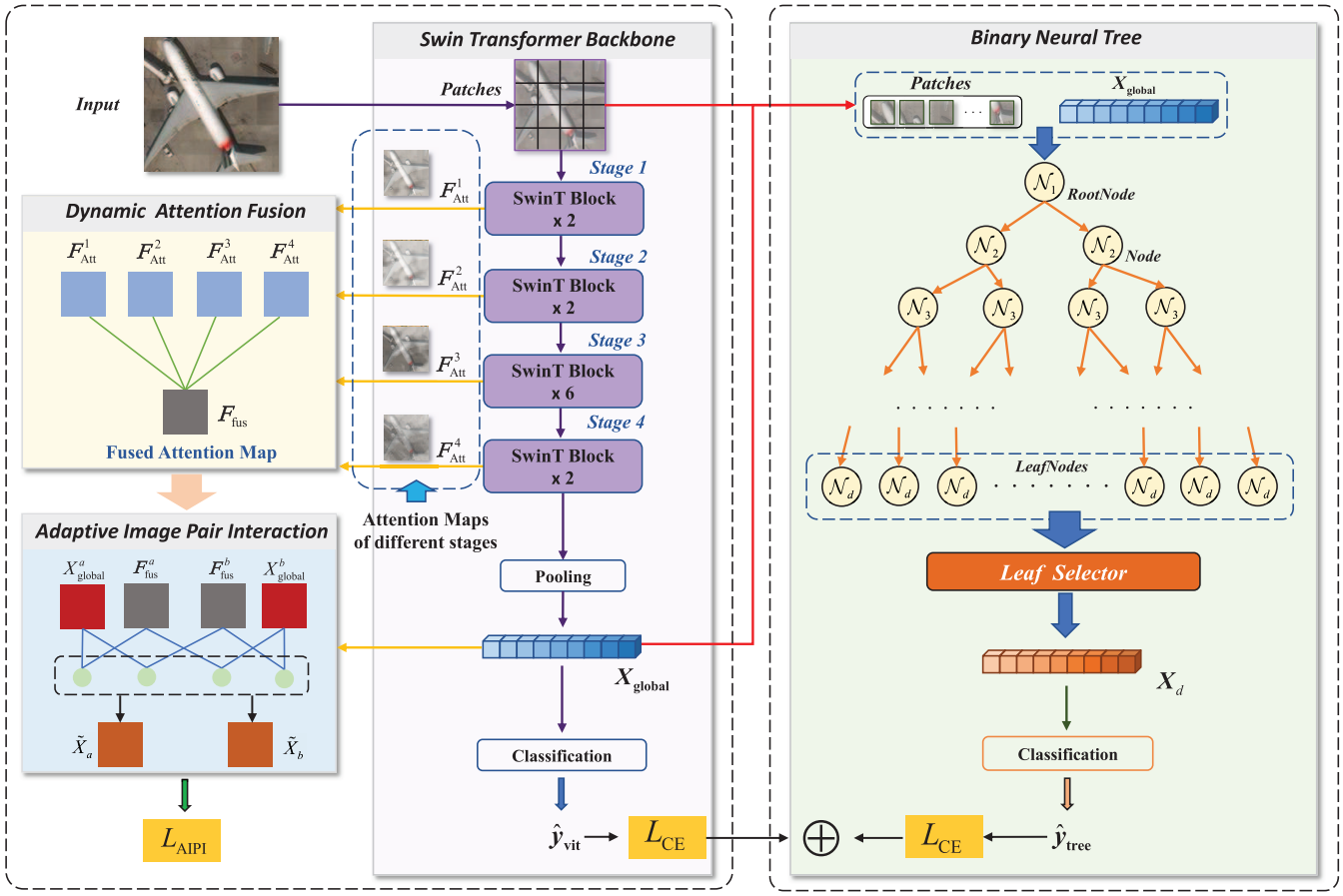


Fig. 3. Overall architecture of our network.

is first divided into a sequence of patches, which are then processed through the four stages. Each stage produces an attention map F_{Att}, and after the final stage, a global average pooling (GAP) layer aggregates all patch-level features to obtain the global feature representation X_{global}.

First, DAFM adaptively fuses the multistage attention maps by computing their degree of dispersion/concentration, resulting in a fused attention map F_{fus}. Based on the global features X^a_{global} and X^b_{global} extracted from two images and their corresponding attention maps F^a_{fus} and F^b_{fus}, AIPI generates the interaction-enhanced features $\hat{X}_a$ and $\hat{X}_b$. These features are then supervised by the proposed loss function L_{AIPI}, which guides the optimization of the SwinT backbone during training. Note that both DAFM and AIPI are used only during the training phase to mitigate the challenges of small interclass variance and large intraclass variance.

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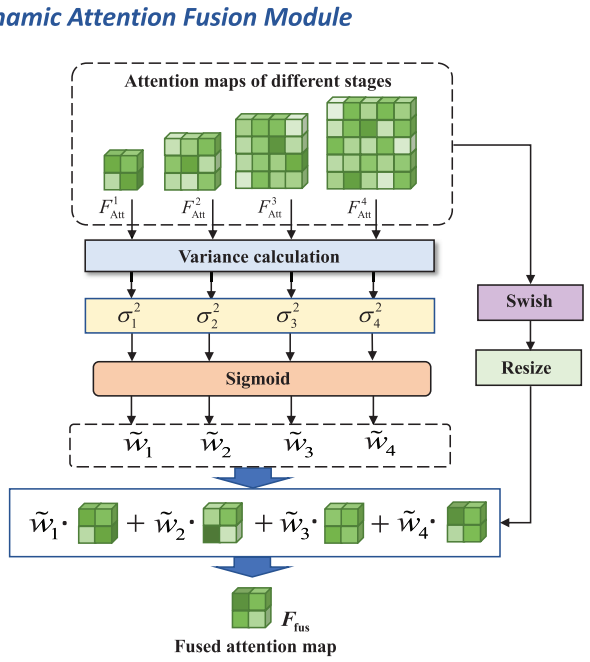


Fig. 4. Illustration of the proposed DAFM.

The BNTM branch takes as input the global feature X_{global} and the patches produced by the SwinT backbone. It iteratively selects the most representative patches in a hierarchical top-down fashion, guided by the current node's feature. This process improves feature expressiveness while simultaneously introducing interpretability: each leaf node in the tree corresponds to a unique decision path, which reflects the specific sequence of regions attended to by the model during classification. During training, we jointly optimize both branches using the sum of cross-entropy losses (L_{CE}) from the SwinT and BNTM outputs. During inference, the final classification result is obtained by averaging the predictions from both branches.

The detailed designs of DAFM, AIPI, and BNTM are presented in Sections III-B–III-D, respectively.

B. Dynamic Attention Fusion Module

In traditional methods, attention map fusion across different stages of ViT architectures is often overlooked or performed through simple summation. To address this, we propose a DAFM, which adaptively adjusts fusion weights based on the dispersion degree of attention maps, enabling more effective feature aggregation.

As illustrated in Fig. 4, DAFM first computes the variance of attention maps at each SwinT stage, capturing their dispersion. Specifically, low-variance (low dispersion) attention maps indicate that the attention weight distribution is more uniform, meaning the model does not have clear focal points. This can lead to weaker concentration of attention, reducing the model's ability to distinguish key regions from the background effectively. On the other hand, high-variance (high dispersion) attention maps exhibit significant differences in weight distribution, where certain areas receive much higher attention than others. This suggests that the model is focusing more on key local features, making such attention maps more suitable for fine-grained classification tasks.

Therefore, to ensure that high-variance attention maps play a greater role in the fusion process while low-variance attention maps contribute less, we use a monotonically increasing sigmoid function to map variance values to fusion weights. This function scales the variance scores into a bounded range, allowing us to interpret them as fusion weights that reflect the relative importance of each stage's attention maps.

The weight $\tilde{w}_i$ for the ith stage is computed based on the spatial variance of its attention map

$$\tilde{w}_i = \phi\left(\text{Var}\left(F^i_{\text{Att}}\right)\right) \tag{1}$$

where F^i_{Att} denotes the attention map output from the ith stage of SwinT, where each element indicates the attention value for a patch. $\text{Var}(F^i_{\text{Att}})$ denotes the variance of attention scores in F^i_{Att}, computed as

$$\text{Var}\left(F^i_{\text{Att}}\right) = \frac{1}{M}\sum_{j=1}^{M}\left(a_j - \mu^i\right)^2, \quad \mu^i = \frac{1}{M}\sum_{j=1}^{M} a_j \tag{2}$$

where $\{a_j\}_{j=1}^{M}$ are the elements of F^i_{Att}, and M is the total number of elements. μ^i is the mean value of F^i_{Att}. $\phi(\cdot)$ is the sigmoid activation function

$$\phi(x) = \frac{1}{1+e^{-x}}. \tag{3}$$

To further enhance the effectiveness of feature fusion, we apply a nonlinear mapping to the attention maps $\{F^i_{\text{Att}}\}$, utilizing the Swish function $\psi(\cdot)$ as the nonlinear transformation. The specific formulation is as follows:

$$\tilde{F}^i_{\text{Att}} = \psi\left(F^i_{\text{Att}}\right) = F^i_{\text{Att}} \cdot \frac{e^{F^i_{\text{Att}}} - 1}{e^{F^i_{\text{Att}}} + 1}. \tag{4}$$

Finally, we perform weighted fusion using the nonlinear features and their corresponding weights. Note that, since the attention maps from the four stages of the SwinT network differ in size, we resize them to match the dimensions of the attention map from the final stage before fusion. The fused attention map F_{fus} is calculated as follows:

$$F_{\text{fus}} = \sum_{i=1}^{4} \text{Resize}\left(F^i_{\text{Att}}\right) \cdot \tilde{w}_i \tag{5}$$

where $\text{Resize}(\cdot)$ is the downsampling operation implemented via max pooling.

This dynamic fusion approach allows the model to adaptively balance the contributions of different attention maps, ensuring that both coarse and fine-grained information are effectively utilized. This improves the model's ability to capture complex visual patterns, particularly in tasks like fine-grained classification, where both global structures and subtle details are crucial for accurate predictions.

C. AIPI Module

To address the challenges of large intraclass variance and small interclass variance, we aim for the network to explore the relationships between the features of image pairs. Inspired

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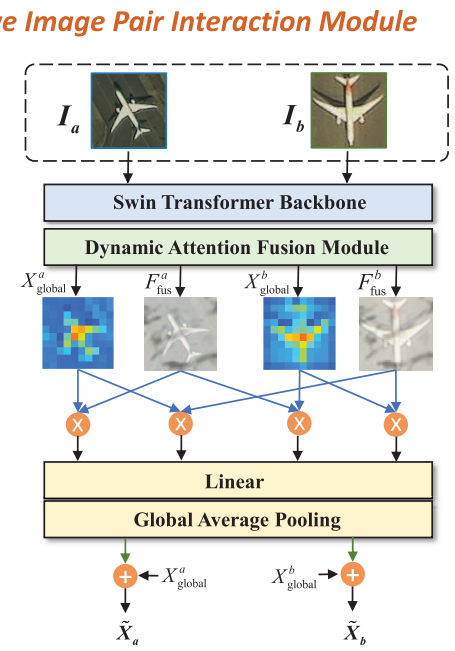


Fig. 5. Illustration of the proposed AIPI.

by previous works [37], [59], we designed the AIPI, which is shown in Fig. 5. This module employs an adaptive matching mechanism that dynamically selects different interaction strategies based on the intraclass or interclass similarity, thereby enhancing the model's robustness.

First, consider the input image pair I_a and I_b. After passing through the backbone, we obtain the global features X^a_{global} and X^b_{global} extracted by the SwinT backbone, along with their corresponding fused attention maps F^a_{fus} and F^b_{fus} produced by the DAFM. Specifically, X^a_{global} and X^b_{global} are obtained after the fourth stage of the SwinT backbone, followed by a GAP layer. The fourth stage contains the deepest and semantically richest features, capturing high-level representations that are more invariant to visual variations such as pose, scale, and background clutter—characteristics critical to fine-grained classification tasks. By selecting features from this layer, we ensure that the subsequent pairwise interaction in the AIPI module is based on semantically meaningful representations, which strengthens the model's ability to reason about subtle differences and similarities between samples.

To refine and enhance the most informative regions during the feature interaction process, we define a feature interaction function $f(X, F)$. This function operates on both the global features and attention maps to perform element-wise modulation and extract the most critical information for each image, which is formulated as

$$f(X, F) = \text{GAP}\left(\text{Linear}\left(X \odot F\right)\right) \tag{6}$$

where $\odot$ represents the element-wise multiplication, followed by a linear layer and a GAP layer. This process emphasizes the key feature areas, helping the model focus on the most informative parts of the images.

Next, we introduce a dynamic matching function $\mathcal{S}(X^a_{\text{global}}, X^b_{\text{global}})$, which selects different interaction strategies based on the intraclass or interclass similarity

$$\mathcal{S}\left(X^a_{\text{global}}, X^b_{\text{global}}\right) = \phi\left(\text{CosSim}\left(X^a_{\text{global}}, X^b_{\text{global}}\right)\right). \tag{7}$$

Here, CosSim represents the cosine similarity between the features X^a_{global} and X^b_{global}, defined as

$$\text{CosSim}\left(X^a_{\text{global}}, X^b_{\text{global}}\right) = \frac{X^a_{\text{global}} \cdot X^b_{\text{global}}}{\left\|X^a_{\text{global}}\right\| \left\|X^b_{\text{global}}\right\|} \tag{8}$$

where $X^a_{\text{global}} \cdot X^b_{\text{global}}$ denotes the dot product between the two features, and $\left\|X^a_{\text{global}}\right\|$ and $\left\|X^b_{\text{global}}\right\|$ denote the L_2 norms of the respective features. $\phi(\cdot)$ is the sigmoid function, which maps the similarity score to the range $[0, 1]$.

Based on the categories of I_a and I_b, and the value of $\mathcal{S}(X^a_{\text{global}}, X^b_{\text{global}})$, we propose a dynamic interaction strategy to adapt to intraclass and interclass variations, enhancing the model's robustness.

Specifically, we denote the refined features after interaction as $\tilde{X}_a$ and $\tilde{X}_b$. The strategy is as follows.

When I_a and I_b belong to the same class:

1) $\mathcal{S}(X^a_{global}, X^b_{global}) > \tau$ *(Intraclass Features With Small Variance):* Enhance the most informative regions within each image without feature interaction

$$\begin{aligned}\tilde{X}_a &= f\left(X^a_{\text{global}}, F^a_{\text{fus}}\right) + X^a_{\text{global}} \\ \tilde{X}_b &= f\left(X^b_{\text{global}}, F^b_{\text{fus}}\right) + X^b_{\text{global}}.\end{aligned} \tag{9}$$

2) $\mathcal{S}(X^a_{global}, X^b_{global}) \leq \tau$ *(Intraclass Features With Large Variance):* Perform feature interaction to handle significant intraclass diversity

$$\begin{aligned}\tilde{X}_a &= f\left(X^a_{\text{global}}, F^b_{\text{fus}}\right) + X^a_{\text{global}} \\ \tilde{X}_b &= f\left(X^b_{\text{global}}, F^a_{\text{fus}}\right) + X^b_{\text{global}}.\end{aligned} \tag{10}$$

When I_a and I_b belong to different classes:

1) $\mathcal{S}(X^a_{global}, X^b_{global}) > \tau$ *(Interclass Features With Small Variance):* Perform mixed feature interaction to emphasize subtle differences

$$\begin{aligned}\tilde{X}_a &= f\left(X^a_{\text{global}}, F^b_{\text{fus}}\right) + X^a_{\text{global}} \\ \tilde{X}_b &= f\left(X^b_{\text{global}}, F^a_{\text{fus}}\right) + X^b_{\text{global}}.\end{aligned} \tag{11}$$

2) $\mathcal{S}(X^a_{global}, X^b_{global}) \leq \tau$ *(Intraclass Features With Large Variance):* No feature interaction; focus on enhancing key regions within each image

$$\begin{aligned}\tilde{X}_a &= f\left(X^a_{\text{global}}, F^a_{\text{fus}}\right) + X^a_{\text{global}} \\ \tilde{X}_b &= f\left(X^b_{\text{global}}, F^b_{\text{fus}}\right) + X^b_{\text{global}}.\end{aligned} \tag{12}$$

Where + is the element-wise addition, and the threshold τ is set to 0.5, as validated in ablation experiments Section IV-D.3.

The training strategy for the AIPI module involves randomly selecting two images at a time from the batch to form a pair for interaction, continuing this process until all images in the batch have been selected. The resulting features from these pairs are then fed into the classifier within the SwinT backbone, where they are optimized jointly with the features X_{global} derived solely from the SwinT backbone using the loss function L_{AIPI} (introduced in Section III-E). It is important

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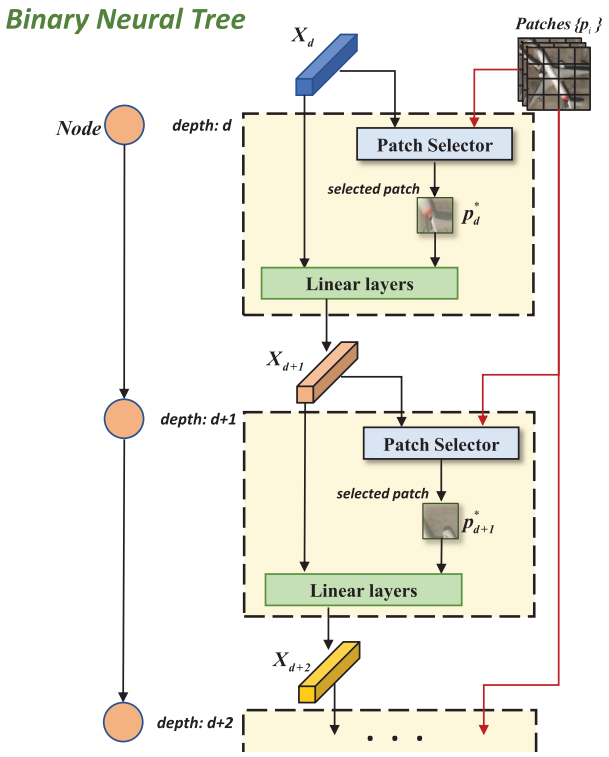


Fig. 6. Illustration of the proposed BNTM.

to note that the AIPI module is only utilized during training and does not participate in the inference stage. Its impact is retained in the optimized backbone parameters, which encode more discriminative and robust features learned through interclass and intraclass interaction and are directly used during inference.

This module utilizes the dynamic matching function $\mathcal{S}(X^a_{\text{global}}, X^b_{\text{global}})$ to select different interaction strategies based on similarity, enhancing the model's adaptability to both intraclass and interclass variations. In addition, it preserves the original features to ensure a comprehensive integration of global and local information, thereby enriching the expressiveness of the feature representation.

D. Binary Neural Tree Module

To enhance the model's ability to adaptively focus on key regions and improve interpretability, we propose a BNTM. Inspired by the hierarchical decision-making process in neural tree structures from previous works [60], [61], [62], BNTM progressively selects and transmits the most informative image patches at each layer of the tree, refining feature representation in a structured manner.

BNTM adopts a binary tree structure, where each node, except for leaf nodes, has two child nodes. Feature transmission occurs from the parent node to its two child nodes, forming a hierarchical, top-down propagation mechanism. The feature transmission between the parent and child nodes is illustrated in Fig. 6. Taking the root node as an example, the input to BNTM includes the global feature X_d extracted by the SwinT backbone and a set of image patches p_i obtained from the original image. The Patch Selector, based on X_d, identifies the most informative patch p^*_d, which is then combined with X_d and passed through a linear transformation layer to generate the feature representation for the next layer, denoted as X_{d+1}. This new feature serves as the parent node in the subsequent layer and continues to propagate downward until the leaf nodes are reached.

As shown in Fig. 3, the nodes at the lowest layer of the binary neural tree are referred to as leaf nodes (which have no child nodes). Assuming the tree has a depth of D, there are a total of 2^D leaf nodes, with each leaf node having its own specific feature representation $\{X^1, \ldots, X^{2^D}\}$. In our method, a tree depth of 6 is selected, as validated in ablation experiments Section IV-D.2. In addition, each leaf node corresponds to a specific feature propagation path from the root to that particular leaf, representing the image patches selected at each layer. Finally, the leaf selector identifies the most optimal leaf node, whose feature representation X serves as the final output of BNTM for classification decisions.

By leveraging explicit path selection, BNTM not only improves classification accuracy but also enhances model interpretability, allowing for a clear visualization of the regions attended to at different hierarchical levels and their contributions to the final classification decision. Further visualization analysis is provided in Section IV-E.

We will detail each module of the BNTM below.

1) Patch Selector: It leverages a multihead attention [63] mechanism to select the most informative patch from a set of image patches based on the parent node's feature representation.

Given a set of image patches $P = \{p_1, p_2, \ldots, p_n\} \in \mathbb{R}^{n\times d}$ and a parent feature $X_d \in \mathbb{R}^{1\times d}$, we compute the attention weights W by applying multihead attention

$$W = \text{MultiHead}\,(X_d, P, P)\,. \tag{13}$$

Specifically, the multihead attention mechanism is defined as

$$\text{MultiHead}\,(Q, K, V) = \text{Concat}\,(h_1, \ldots, h_H)\,W^O \tag{14}$$

$$h_i = \text{Attention}_i\,(Q, K, V) = \text{Softmax}\left(\frac{QW_i^Q\left(KW_i^K\right)^\top}{\sqrt{d_k}}\right)VW_i^V \tag{15}$$

where $Q = X_d$ is the query (parent node feature), and $K = V = P$ are the key and value (patch set). H denotes the number of attention heads. $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d\times d_k}$ are learnable projection matrices for the ith head, and $W^O \in \mathbb{R}^{Hd_k\times d}$ is the output projection matrix. The resulting attention weights $W \in \mathbb{R}^{1\times n}$ indicate the relevance of each patch to the parent node feature.

We then select the most informative patch by identifying the index corresponding to the maximum weight

$$i = \text{argmax}_i\,(W) \tag{16}$$

where $\text{argmax}_i(W)$ represents selecting the index i corresponding to the patch with the highest attention score. Thus, the

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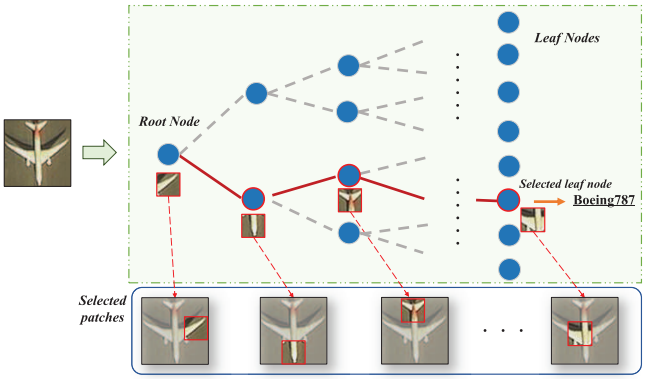


Fig. 7. Illustration of the interpretability provided by BNTM. At each node along the decision path, the model focuses on a discriminative region of the aircraft, such as the nose, wings, fuselage, and tail. These progressively selected patches form a clear, interpretable reasoning path from root to leaf, revealing how the model arrives at its final classification.

selected patch $p_d^* = p_i$ acts as the key information carrier and is forwarded to the next step.

2) Linear Layers: First, the global feature X_d from the parent node and the selected patch p_d^* are transformed to the same dimensionality. Then, the two feature vectors are concatenated and passed through two fully connected layers with a nonlinear activation function to further optimize them, generating the updated feature representation for the child node

$$X_{d+1} = \mathrm{FC}_2\left(\mathrm{ReLU}\left(\mathrm{FC}_1\left(\mathrm{Concat}\left(X_d, p_d^*\right)\right)\right)\right). \tag{17}$$

Here, Concat$(\cdot,\cdot)$ denotes the operation of concatenating two feature vectors, FC_1 and FC_2 are two fully connected layers, and ReLU$(\cdot)$ is the nonlinear activation function.

3) Leaf Selector: It is responsible for selecting the most informative leaf node from all leaf nodes at the final layer of the binary tree. Specifically, it takes the feature representations of all leaf nodes $\{X^1, \ldots, X^{2^D}\}$ as input and computes a weight distribution for each using a fully connected layer. The weights are then normalized using the softmax function to represent the importance of each leaf node

$$W_x = \mathrm{Softmax}\left(\mathrm{FC}\left(\left\{X^i\right\}_{i=1}^{2^D}\right)\right) \tag{18}$$

where D is the depth of the tree. And the index i of the leaf node with the highest weight is selected by

$$i = \mathrm{argmax}_i\left(W_x\right). \tag{19}$$

Finally, the corresponding feature X^i is extracted as the final output for classification, serving as the decision of the tree.

As illustrated in Fig. 7, at each node in the binary tree, BNTM selects a patch that corresponds to a specific, spatially meaningful region in the original image. Once a leaf node is selected, a unique path from the root to that leaf is determined, and the final classification is based on the features aggregated along this path. This path thus serves as an explicit reasoning trace for the model's decision.

The hierarchical patch selection mechanism enables us to answer key interpretability questions such as "*Where is the model looking?*" and "*How does it make decisions step by step?*" in a transparent and structured way. By visualizing the selected patches at each tree layer, users can clearly understand how the model progressively focuses on increasingly discriminative regions, ultimately enhancing the interpretability of the classification process.

E. Loss Function

1) Tree Loss: The first part is the joint optimization of both the tree and ViT-based decisions. Specifically, this loss is the sum of the cross-entropy loss functions from the predictions made by the SwinT backbone and the BNTM, ensuring that both components are optimized in tandem

$$L_1 = \mathrm{CrossEntropy}\left(\hat{y}_{\mathrm{vit}}, y\right) + \mathrm{CrossEntropy}\left(\hat{y}_{\mathrm{tree}}, y\right) \tag{20}$$

where $\hat{y}_{\mathrm{vit}}$ and $\hat{y}_{\mathrm{tree}}$ denote the predicted class probability distributions produced by the SwinT backbone and BNTM, respectively, and y denotes the ground-truth label.

The cross-entropy loss is defined as

$$\mathrm{CrossEntropy}\left(\hat{y}, y\right) = -\sum_{i=1}^{C} y_i \log\left(\hat{y}_i\right) \tag{21}$$

where C is the total number of classes, y_i is the one-hot encoded ground-truth label (i.e., $y_i = 1$ if the sample belongs to class i, and $y_i = 0$ otherwise), and $\hat{y}_i$ denotes the predicted probability for class i.

2) AIPI Loss: To address the issue of small interclass variance and large intraclass variance, we designed the L_{AIPI} loss, which is composed of three parts: intraclass consistency loss L_{intra} and interclass separability loss L_{inter}.

Specifically, for all randomly selected image pairs N within a batch, let $\tilde{X}_a$ and $\tilde{X}_b$ represent the feature embeddings of images I_a and I_b after passing through the backbone and AIPI.

For image pairs belonging to the same class, we want the network to pull their feature embeddings closer in the feature space to enhance the robustness of intraclass features. This is achieved by minimizing the Euclidean distance between the feature representations of intraclass image pairs

$$L_{\mathrm{intra}} = \frac{1}{N} \sum_{(I_a, I_b) \in \mathrm{intraclass}} \left\| \tilde{X}_a - \tilde{X}_b \right\|^2. \tag{22}$$

For image pairs from different classes, we aim for the network to push their feature embeddings farther apart in the feature space, enhancing the distinctiveness of interclass features. This is achieved by maximizing the Euclidean distance between the feature representations of interclass image pairs

$$L_{\mathrm{inter}} = -\frac{1}{N} \sum_{(I_a, I_b) \in \mathrm{interclass}} \left\| \tilde{X}_a - \tilde{X}_b \right\|^2. \tag{23}$$

So, the L_{AIPI} is defined as

$$L_{\mathrm{AIPI}} = L_{\mathrm{inter}} + L_{\mathrm{intra}}. \tag{24}$$

And the overall loss is then defined as

$$L = L_1 + L_{\mathrm{AIPI}}. \tag{25}$$

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TABLE I
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE FAIR1M-1.0 DATASET

Class	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	Total
Training	3949	1673	1532	1669	6056	2504	1599	1064	166	135	**20347**
Testing	987	418	383	425	1514	626	400	266	42	34	**5095**

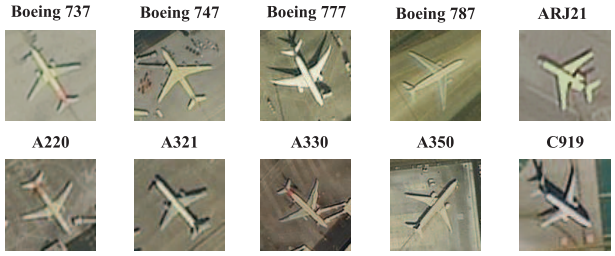


Fig. 8. Examples of data samples from the FAIR1M-1.0 dataset.

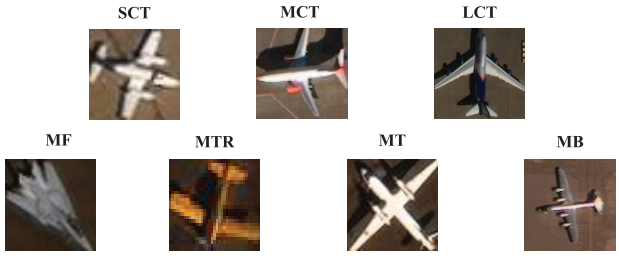


Fig. 9. Examples of data samples from the RarePlanes dataset.

TABLE II
DISTRIBUTION OF TRAINING AND TESTING SAMPLES FOR EACH CLASS IN THE RAREPLANES

Class	LCT	MCT	SCT	MF	MTR	MT	MB	Total
Training	1008	5440	9779	173	17	253	6	**16676**
Testing	661	2379	2472	97	9	177	4	**5779**

IV. EXPERIMENTS AND RESULTS

In this section, we evaluate the effectiveness of our approach. Section IV-A introduces the experimental setup, including datasets, implementation details, and metrics. Section IV-B compares our method with state-of-the-art approaches. Section IV-C analyzes the confusion matrix. Section IV-D presents ablation studies. Section IV-E demonstrates interpretability and visualization results. Finally, Section IV-F reports the computational cost.

A. Experimental Setups

1) Dataset Descriptions: The FAIR1M-1.0 [39] dataset is a high-resolution remote sensing image benchmark for object detection, consisting of images with resolutions ranging from 0.3 to 0.8 m collected from various platforms. It covers five major categories and 37 subcategories. For our experiments, we selected ten subcategories from the aircraft category: Boeing 737 (B737), Boeing 747 (B747), Boeing 777 (B777), Boeing 787 (B787), Airbus A320 (A321), Airbus A220 (A220), Airbus A330 (A330), Airbus A350 (A350), COMAC C919 (C919), and COMAC ARJ21 (ARJ21), with sample images shown in Fig. 8. We split the data into training and testing sets in a 4:1 ratio, and the sample distribution for each class is provided in Table I. Notably, the dataset predominantly consists of Boeing and Airbus subcategories, which exhibit subtle interclass differences. In addition, as shown in Table I, there is an imbalance in the number of training samples across classes. These characteristics make the dataset well-suited for evaluating our method's ability to address these issues.

RarePlanes [40] is a high-resolution dataset specifically designed for aircraft detection in remote sensing images, containing a total of 22 475 aircraft instances. The dataset is categorized into seven classes: small civil transport (SCT), medium civil transport (MCT), large civil transport (LCT), military fighter (MF), military bomber (MB), military transport (MT), and military trainer (MTR). Examples of these classes are shown in Fig. 9. We selected 75% of the data for training and 25% for testing, with the number of samples per class listed in Table II. As shown, the dataset exhibits significant class imbalance, posing considerable challenges for both model training and evaluation.

2) Implementation Details: All experiments are conducted using the PyTorch framework on a Linux server equipped with four NVIDIA RTX 3090 GPUs. We adopt the officially released ImageNet-21k pretrained weights for the SwinT backbone. During training, all input images were resized and cropped to a fixed resolution of 224×224.

For optimization, we used the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay of 0.05, and an epsilon of 1×10^{-8}. The base learning rate was set to $5e^{-4}$, with a warm-up learning rate of $5e^{-7}$ for the first 5 epochs. A cosine annealing scheduler was adopted to gradually reduce the learning rate to a minimum of $5e^{-6}$ during training. The weight decay was set to 0.05, and gradient clipping with a maximum norm of 5.0 was applied to stabilize training. The model was trained for 120 epochs on the FAIR1M1.0 dataset and for 150 epochs on the RarePlanes dataset.

3) Evaluation Metrics: In our experiments, we primarily employ two evaluation metrics: top-1 accuracy and overall accuracy (OA).

4) Top-1 Accuracy: This metric calculates the proportion of samples for which the top predicted label matches the ground truth label. It reflects the model's ability to correctly identify the most probable class

$$\text{Top-1 Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \tag{26}$$

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TABLE III
COMPARISONS OF TOP-1 ACCURACY FOR EACH CATEGORY AND OA WITH THE SOTA FINE-GRAINED METHODS ON FAIR1M-1.0 DATASET

Method	Top-1 Accuracy (%)										OA (%)
	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	
PMG [64] (TPAMI'22)	55.40	78.27	53.82	74.83	63.56	75.42	79.41	72.61	73.58	66.84	68.12
MCPNet [65] (CVPR'24)	57.88	83.93	55.12	76.46	65.34	79.76	76.29	74.48	71.34	75.41	71.70
FS-FGIC [66] (AAAI'24)	63.73	87.54	58.73	72.55	60.34	76.18	78.70	78.75	65.18	81.94	73.26
DPSIS [27] (TCYB'22)	65.73	89.83	60.73	75.12	62.89	79.53	81.16	81.19	70.76	74.48	74.45
PETDet [43] (TGRS'23)	62.46	86.41	62.49	77.39	66.12	78.45	82.53	84.34	68.76	68.43	73.44
LSDA-Net [67] (TIP'24)	60.21	90.12	60.68	71.68	62.67	76.82	78.81	78.82	73.51	72.94	71.53
BIDE-Net [68] (TPAMI'24)	66.82	91.35	63.25	76.21	64.45	75.67	83.34	85.56	72.39	71.85	75.92
SFRNet [49] (TGRS'23)	67.57	84.76	64.91	74.90	67.23	77.34	84.62	84.43	77.82	74.35	76.65
MSRIP-Net [38] (TGRS'24)	68.82	96.37	64.78	82.74	73.82	85.73	92.38	90.71	84.21	84.44	80.74
APPEAR [12] (TCYB'24)	64.92	90.48	61.27	80.65	70.49	81.29	89.45	88.27	81.36	84.24	76.83
TransFG [34] (AAAI'22)	66.74	91.12	58.22	80.35	71.94	82.43	86.17	84.32	69.45	78.19	76.21
TransIFC [69] (TMM'23)	60.15	84.28	63.45	71.98	67.13	76.40	78.93	77.68	71.02	72.83	73.01
IELT [70] (TMM'23)	69.12	92.03	60.01	79.26	63.84	74.21	84.32	82.05	77.38	80.12	76.84
PMRC [71] (CVPR'23)	62.83	86.35	64.82	76.19	66.59	80.07	81.45	79.23	72.94	77.36	75.12
Mix-ViT [35] (PR'23)	59.45	89.22	61.04	74.88	64.77	77.90	83.01	80.67	70.01	74.42	73.26
SEB [72] (TPAMI'23)	65.26	92.15	60.83	77.26	70.15	79.31	81.65	76.58	76.30	77.52	75.11
EFM-Net [44] (TGRS'23)	67.38	89.93	62.75	78.44	69.21	81.67	85.62	82.10	73.26	76.83	76.50
An-Eru [73] (CVPR'23)	64.11	87.62	60.89	75.13	65.70	79.03	82.92	81.47	72.58	75.96	74.52
MP-FGVC [74] (AAAI'24)	61.78	85.49	65.34	72.66	67.42	78.11	79.47	77.85	71.00	73.02	72.83
FET-FGVC [37] (PR'24)	70.22	92.15	66.93	80.73	72.84	84.05	86.98	85.17	75.80	79.60	79.46
ACC-ViT [48] (PR'24)	62.97	88.30	61.82	74.40	68.09	80.76	83.23	80.22	74.94	76.17	74.70
MFAL [75] (TGRS'24)	65.44	90.04	64.18	76.92	70.11	83.28	85.77	83.64	76.85	78.34	76.31
VMamba [76] (ArXiv'24)	63.73	86.81	59.76	73.89	66.47	77.42	81.39	80.71	71.96	74.00	73.01
MetaIRNet [77] (TPAMI'24)	68.36	90.58	63.91	79.85	69.93	82.04	86.43	83.22	75.60	77.15	77.19
GCL [78] (TIP'25)	66.58	91.77	62.14	78.93	70.36	80.65	84.19	82.87	76.13	75.49	76.08
Ours	**83.84**	**99.10**	**72.42**	**88.64**	**83.42**	**90.26**	**96.75**	**93.84**	**87.49**	**86.15**	**88.01**

where N_{correct} is the number of correctly classified samples, and N_{total} is the total number of test samples.

5) Overall Accuracy: This metric measures the overall proportion of correctly classified samples across all test samples

$$\text{OA} = \frac{\sum_{c=1}^{C} N_{\text{correct}}^{(c)}}{\sum_{c=1}^{C} N_{\text{total}}^{(c)}} \tag{27}$$

where C is the total number of categories, and $N_{\text{correct}}^{(c)}$ and $N_{\text{total}}^{(c)}$ denote the number of correctly classified and total samples for category c, respectively.

B. Comparisons With the State-of-the-Art Methods

1) Results on FAIR1M-1.0: Table III reports the top-1 accuracy across aircraft categories and OA on the FAIR1M-1.0 dataset. Our method consistently outperforms existing fine-grained classification approaches across all categories.

Notably, in challenging classes with high intraclass variance and small interclass differences (e.g., B737, B777, and ARJ21), our method achieves accuracies of 83.84%, 72.42%, and 87.49%, respectively—substantially surpassing previous best results. It also reaches 99.10% on B747 and 96.75% on A330, the highest among all methods.

In terms of OA, our model achieves an OA of 88.01%, clearly outperforming top-performing baselines such as MSRIP-Net (80.74%) and FET-FGVC (79.46%). These results highlight the strength of our framework in extracting fine-grained discriminative cues and delivering robust performance across diverse aircraft categories.

2) Results on RarePlanes: Table IV summarizes classification results on the RarePlanes dataset. Our method achieves the highest top-1 accuracy and OA compared with all existing methods.

For well-represented categories such as SCT and LCT, our method reaches 96.58% and 97.62%, exceeding the closest baselines by over 1–2 percentage points. In more difficult categories such as MF and MTR, where data scarcity and high intraclass variance pose challenges, our approach still performs robustly—achieving 91.37% and 88.89%, respectively.

A particularly challenging class is MB, which contains only ten samples. Due to the extremely limited test data, many methods yield unstable performance (0%–50%). Our model correctly classifies all test images, achieving 100.00% accuracy.

Overall, our method achieves an OA of 94.74%, substantially outperforming MSRIP-Net (92.38%) and APPEAR (89.45%), confirming the effectiveness and generalizability of our approach in diverse and low-data remote sensing scenarios.

C. Analysis of the Confusion Matrices

As shown in Fig. 10, we compare the confusion matrix of our proposed method with those of the two next-best methods, MSRIP-Net and APPEAR, on the RarePlanes (top) and FAIR1M-1.0 (bottom) datasets. This analysis highlights the strengths of our approach, especially in handling complex categories with fine-grained distinctions.

1) Confusion Matrices of RarePlanes: Our method consistently outperforms MSRIP-Net and APPEAR across all categories in the RarePlanes dataset, as shown in Fig. 10. In

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TABLE IV

COMPARISONS OF TOP-1 ACCURACY FOR EACH CATEGORY AND OA WITH THE SOTA FINE-GRAINED METHODS ON RAREPLANES DATASET

Method	Top-1 Accuracy (%)							OA (%)
	SCT	MCT	LCT	MF	MB	MT	MTR	
PMG [64] (TPAMI'22)	85.91	78.02	83.20	64.39	0	75.20	11.11	78.93
MCPNet [65] (CVPR'24)	87.20	73.62	85.03	66.29	0	78.27	22.22	81.15
FS-FGIC [66] (AAAI'24)	83.74	77.63	88.74	62.85	25.00	76.20	11.11	83.42
DPSIS [27] (TCYB'22)	87.28	79.38	93.85	67.74	0	78.67	56.56	84.36
PETDet [43] (TGRS'23)	82.36	76.98	82.94	67.09	0	78.02	33.33	83.67
LSDA-Net [67] (TIP'24)	80.74	80.02	80.48	61.23	25.00	76.27	66.67	81.93
BIDE-Net [68] (TPAMI'24)	86.84	81.73	83.24	66.24	0	85.76	66.67	85.17
SFRNet [49] (TGRS'23)	91.35	82.33	88.09	69.94	25.00	83.84	33.33	86.25
MSRIP-Net [38] (TGRS'24)	95.28	89.62	95.52	89.25	75.00	94.28	88.89	92.38
APPEAR [12] (TCYB'24)	93.12	86.54	92.28	70.83	50.00	86.36	66.67	89.45
TransFG [34] (AAAI'22)	92.30	84.20	91.45	68.02	25.00	84.32	66.67	88.21
TransIFC [69] (TMM'23)	90.12	81.87	89.65	66.35	50.00	81.93	66.67	86.44
IELT [70] (TMM'23)	91.84	83.22	90.53	68.58	25.00	82.66	66.67	87.62
PMRC [71] (CVPR'23)	92.01	82.17	89.23	67.01	50.00	83.25	66.67	87.12
Mix-ViT [35] (PR'23)	90.78	80.89	88.60	66.75	25.00	82.87	66.67	86.02
SEB (TPAMI'23) [72]	93.32	86.35	92.14	67.94	25.00	87.36	66.67	88.26
EFM-Net [44] (TGRS'23)	92.45	83.49	90.91	67.94	25.00	83.44	66.67	87.54
An-Eru [73] (CVPR'23)	89.78	80.92	87.13	65.17	25.00	80.64	66.67	85.24
MP-FGVC [74] (AAAI'24)	88.35	79.87	86.70	66.14	50.00	80.91	66.67	84.86
FET-FGVC [37] (PR'24)	94.67	88.02	94.24	88.21	50.00	93.38	88.89	91.55
ACC-ViT [48] (PR'24)	90.35	81.38	89.34	65.86	50.00	82.11	66.67	85.91
MFAL [75] (TGRS'24)	91.24	83.15	90.62	67.43	50.00	83.02	66.67	87.26
VMamba [76] (ArXiv'24)	88.97	80.46	86.57	64.89	50.00	80.53	66.67	84.62
MetaIRNet [77] (TPAMI'24)	93.05	85.29	92.01	68.81	75.00	85.40	88.89	89.93
GCL [78] (TIP'25)	91.75	84.48	90.63	67.63	50.00	83.76	66.67	87.62
Ours	**96.58**	**92.40**	**97.62**	**91.37**	**100.00**	**96.35**	**88.89**	**94.74**

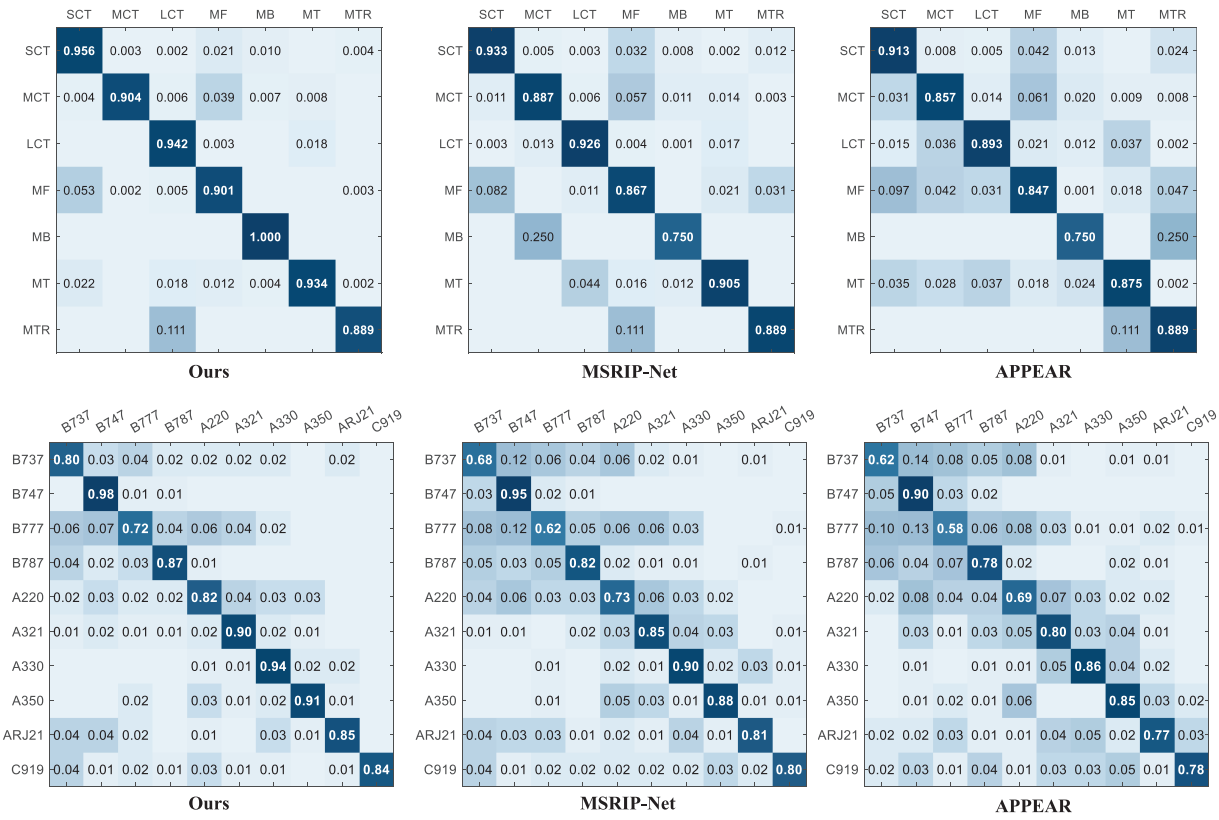


Fig. 10. Confusion matrices of ours, MSRIP-Net, and APPEAR methods on RarePlanes (top) and FAIR1M-1.0 (bottom) datasets.

SCT, MCT, and LCT, it achieves 95.6%, 90.4%, and 94.2% accuracy, respectively, showing reduced interclass confusion and better feature distinction. For the more challenging MF category, our method reaches 90.1%, significantly surpassing

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TABLE V
ABLATION EXPERIMENTS RESULTS ON THE FARI1M-1.0 DATASET

No.	Components				Top-1 Accuracy										OA(%)
	SwinT	DAFM	AIPI	BNTM	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	
1	✓				65.24	82.25	55.63	69.74	68.35	75.38	77.29	74.41	70.33	72.18	73.48
2	✓	✓			70.74	88.36	60.49	77.92	74.31	79.06	82.51	78.46	75.18	76.49	77.26
3	✓		✓		72.84	89.16	62.03	78.28	75.61	79.51	82.18	77.13	75.02	77.93	78.21
4	✓			✓	73.25	88.24	64.39	77.62	76,03	78.83	83.25	79.46	75.28	74.82	78.35
5	✓	✓	✓		78.33	95.35	68.32	84.09	80.57	84.23	91.63	84.97	81.06	82.37	83.81
6	✓	✓	✓	✓	**83.84**	**99.10**	**72.74**	**88.24**	**83.57**	**90.25**	**96.83**	**93.03**	**87.35**	**86.15**	**88.01**

TABLE VI
ABLATION EXPERIMENTS RESULTS ON THE RAREPLANES DATASET

No.	Components				Top-1 Accuracy							OA(%)
	SwinT	DAFM	AIPI	BNTM	SCT	MCT	LCT	MF	MB	MT	MTR	
1	✓				85.35	78.63	84.28	77.35	0.00	77.24	11.11	76.91
2	✓	✓			87.37	82.73	89.35	82.03	50.00	84.26	56.56	83.23
3	✓		✓		88.04	83.91	91.03	83.19	75.00	88.64	**88.89**	84.76
4	✓			✓	90.10	82.94	90.43	83.51	50.00	89.76	**66.67**	84.32
5	✓	✓	✓		92.1	88.13	92.04	87.29	75.00	92.51	**88.89**	89.65
6	✓	✓	✓	✓	**96.65**	**92.58**	**97.43**	**91.62**	**100.00**	**96.73**	**88.89**	**94.74**

MSRIP-Net (69.9%) and APPEAR (66.5%). In the low-shot MB class (four test samples), our method is the only one to achieve 100% accuracy, while MSRIP-Net and APPEAR obtain 75.0% and 50.0%. For MT and MTR, our method attains 93.4% and 88.9%, showing strong generalization and robustness even under limited data conditions.

2) Confusion Matrices of FAIR1M-1.0: On the FAIR1M-1.0 dataset, our approach demonstrates superior discriminative capability, especially in categories with high visual similarity. For example, in the Boeing series, it achieves 80% on B737, 98% on B747, 72% on B777, and 87% on B787, all outperforming MSRIP-Net and APPEAR. Similarly, in the Airbus series, our method yields 82% (A220), 90% (A321), 94% (A330), and 91% (A350), consistently ahead of baselines. For domestic aircraft types ARJ21 and C919, it also maintains leading performance at 85% and 84%, respectively. These results highlight the model's robustness and ability to capture fine-grained distinctions in both high-similarity and data-limited settings.

Overall, our method surpasses MSRIP-Net and APPEAR across most categories, showing superior accuracy and robustness in challenging, data-limited scenarios. Confusion matrix analysis highlights our approach's ability to capture fine-grained distinctions and minimize misclassification, especially within closely related aircraft types on the RarePlanes and FAIR1M-1.0 datasets.

D. Ablative Analysis

1) Analysis of the Effectiveness of the Different Modules: In Tables V and VI, we present the results of ablation experiments conducted on the FAIR1M-1.0 and RarePlanes datasets to assess the contributions of different modules to model performance.

On the FAIR1M-1.0 dataset (see Table V), the baseline SwinT alone yields relatively low accuracy in challenging categories such as B737 (65.24%) and B777 (55.63%). Adding DAFM or AIPI individually leads to notable performance improvements, indicating their effectiveness in enhancing attention quality and feature discrimination, respectively. Notably, the addition of BNTM alone (Row 4) also results in improved accuracy (e.g., +4.87% OA over SwinT), validating its effectiveness even without auxiliary training modules. When all components are combined, the model achieves the best performance across all classes, with nearly 100% accuracy in B747 and significant gains in other fine-grained categories.

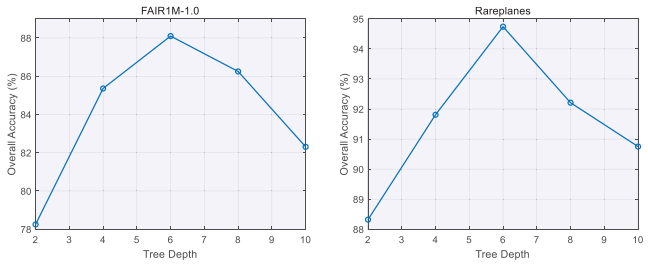


Fig. 11. Effectiveness of the tree depth on FAIR1M-1.0 (left) and RarePlanes (right) datasets.

A similar trend is observed on the RarePlanes dataset (see Table VI). SwinT + BNTM (Row 4) improves over the baseline by 7.41% in OA, especially in categories like MF and MT. Compared to SwinT + DAFM or SwinT + AIPI, this confirms the standalone value of BNTM in learning discriminative structures. The full model further boosts performance, achieving 96.65% and 100% accuracy in SCT and MB, respectively. Ultimately, the combination of all modules yields significant advancements in model performance for the SCT and LCT categories, with SCT achieving 96.65% accuracy and MF reaching 100%, demonstrating the model's robustness and adaptability across various classes.

2) Tree Depth Selection of the Method: Fig. 11 illustrates the impact of varying tree depths on the overall model accuracy. The results indicate that accuracy peaks at a tree depth of 6. While further increasing the depth maintains a relatively

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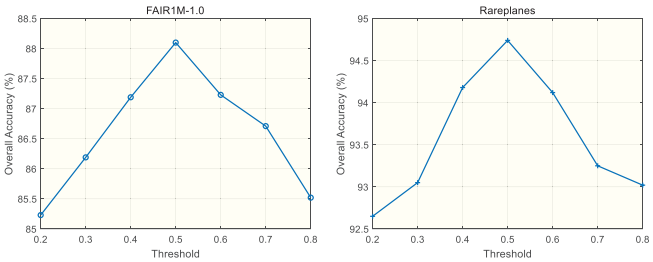


Fig. 12. Effectiveness of the similarity threshold τ on FAIR1M-1.0 (left) and RarePlanes (right) datasets.

high accuracy, there is a noticeable decline beyond a depth of 6. This decline may be attributed to the exponential growth of linear layers as the depth increases, which can lead to overfitting and adversely affect the performance of the final leaf selector. Thus, a tree depth of 6 offers an optimal balance between model complexity and accuracy, making it the most suitable choice for this model.

3) Similarity Threshold τ Selection of the Method: The similarity threshold τ is a crucial parameter in our method, determining the level of similarity required between feature pairs during interaction. Selecting an appropriate τ is essential, as it directly impacts model performance by controlling the balance between interclass separation and intraclass compactness. As shown in Fig. 12, we empirically evaluate various values of τ and observe that a threshold of 0.5 yields the highest OA of 88.10%, effectively maximizing classification performance by enhancing feature discrimination while avoiding overfitting. This optimal threshold is selected based on extensive ablation studies, which confirm that lower values result in reduced accuracy due to under-segmentation of features, while higher values lead to over-segmentation, causing model instability.

E. Interpretability and Visualization Analysis

To validate the interpretability of our method, we first provide a qualitative analysis by visualizing the decision-making process of the model. As shown in Fig. 13, we display the five patches selected along the decision path (excluding the root node), clearly highlighting the spatial regions attended to by the model at each decision level. From the figure, it is evident that our method primarily focuses on the rigid structural components of the aircraft—such as the nose, wings, and fuselage—which aligns with the logical reasoning patterns of human experts who rely on key parts for target classification.

While some selected patches include portions of the background, this is likely due to the complexity and contextual richness of remote sensing imagery. Such behavior suggests that the model may also leverage contextual information during inference, further emphasizing the transparency and traceability of its decision-making process.

To further quantify the interpretability of the proposed BNTM, we introduce a region hit rate (RHR) metric and provide visualizations of the model's decision process. As illustrated in Fig. 13, we employ Chat-GPT to simulate a human expert by identifying five key discriminative regions (e.g., nose, wings, fuselage, and tail) for each aircraft based on its semantic characteristics. We then visually compare these expert-defined regions with the five patches selected along the decision path of BNTM. Since GPT selects regions over the full image while BNTM operates on discrete patches, exact alignment is not always feasible. Therefore, we manually assess whether each selected patch semantically overlaps with any of the GPT-defined regions. A patch is marked as a hit if such semantic overlap is observed; otherwise, it is marked as a miss. The RHR for each image is then computed as the ratio of hits among the five selected patches.

This manual evaluation is performed on 100 randomly selected test images from each dataset. As shown in Table VII, BNTM achieves an RHR of 84.6% on the FAIR1M-1.0 dataset and 81.2% on the RarePlanes dataset, demonstrating that the selected patches are largely consistent with human-understandable discriminative cues, thus validating the interpretability of our method.

TABLE VII
RHR OF BNTM ON TWO DATASETS

Dataset	Test Images	RHR (%)
FAIR1M-1.0	100	84.6
RarePlanes	100	81.2

TABLE VIII
COMPUTATIONAL COST ANALYSIS OF OUR METHODS

Model	Params (M)	GFLOPs	Training Time (h/epoch)	Inference Time (ms/img)
SwinT	28.0	4.5	0.50	3.4
+ DAFM	**29.5** (↑1.5)	**5.1** (↑0.6)	**0.62** (↑0.12)	**4.3** (↑0.9)
+ DAFM + AIPI	**30.2** (↑2.2)	**5.7** (↑1.2)	**0.84** (↑0.34)	**4.3** (↑0.9)
+ DAFM + AIPI + BNTM	**31.2** (↑3.2)	**6.2** (↑1.7)	**0.90** (↑0.40)	**7.6** (↑4.2)

F. Computational Cost Analysis

This section presents the computational cost analysis of our method. We evaluate the impact on model parameters, GFLOPs, training time per epoch, and inference time per image, providing a reference for practitioners.

Table VIII summarizes the computational overhead introduced by each module. Adding DAFM leads to a modest increase in parameters (+1.5M) and GFLOPs (+0.6), with only a slight rise in training and inference time, demonstrating its efficiency. The AIPI module increases training time (by 0.34 h/epoch) due to additional interclass/intraclass interaction supervision but does not affect inference efficiency, as it is only active during training.

Finally, incorporating the BNTM module leads to the most noticeable increase in inference time (from 4.3 to 7.6 ms/img), despite a relatively small increase in parameters (+1.0M). This is due to the nature of BNTM, which performs hierarchical token-level reasoning across multiple layers. Each stage of BNTM involves multihead attention mechanisms and several linear layers, which contribute significantly to runtime during inference, even though the parameter size remains moderate.

As shown in Table IX, our method maintains a moderate level of computational complexity. The parameter count

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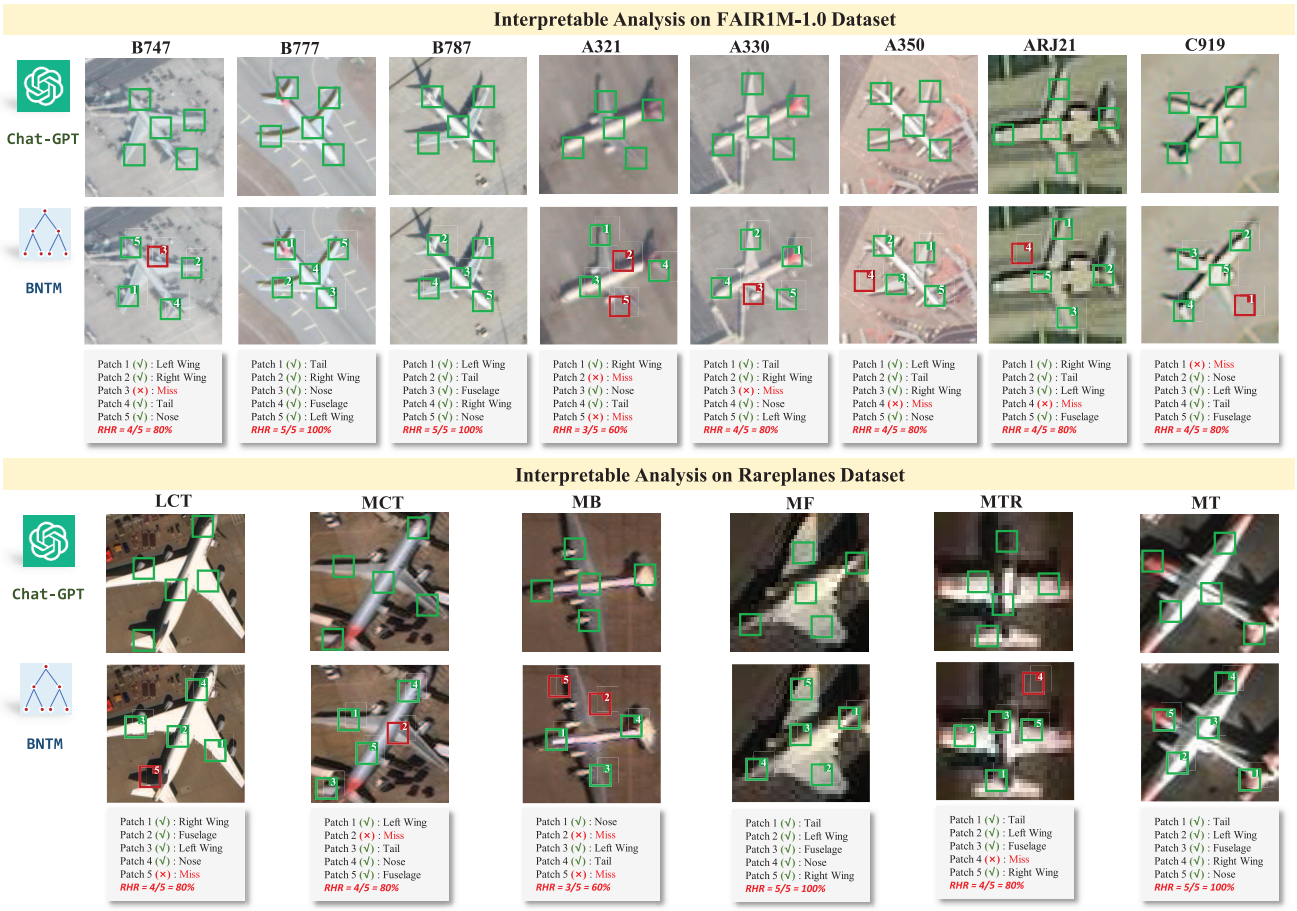


Fig. 13. Comparison between GPT-defined key discriminative regions and BNTM-selected patches for aircraft classification. The top row shows the five discriminative regions selected by ChatGPT simulating a human expert. The bottom row shows the five patches sequentially selected by the BNTM along its decision path. In the BNTM row, green boxes indicate semantically aligned selections ("hit"), while red boxes indicate misaligned or irrelevant regions ("miss"). A summary below each image reports the hit/miss status of each patch, along with the RHR, calculated as the proportion of semantically aligned patches among the five selected.

TABLE IX
COMPUTATIONAL COST ANALYSIS OF DIFFERENT METHODS

Model	Params (M)	GFLOPs	Training Time (h/epoch)	Inference Time (ms/img)
PMG [64] (TPAMI'22)	22.6	4.7	0.6	8.0
MCPNet [65] (CVPR'24)	28.8	10.2	1.1	5.7
FS-FGIC [66] (AAAI'24)	20.7	4.2	0.8	6.3
DPSIS [27] (TCYB'22)	18.2	3.8	0.5	5.5
PETDet [43] (TGRS'23)	46.3	14.3	0.4	10.7
LSDA-Net [67] (TIP'24)	27.3	4.2	0.7	6.5
BIDE-Net [68] (TPAMI'24)	24.6	7.4	0.5	4.3
SFRNet [49] (TGRS'23)	56.4	8.6	0.8	6.4
MSRIP-Net [38] (TGRS'24)	18.3	5.3	0.7	12.3
APPEAR [12] (TCYB'24)	47.6	12.4	1.3	8.8
TransFG [34] (AAAI'22)	25.1	5.8	0.7	7.2
TransIFC [69] (TMM'23)	29.5	6.5	0.8	6.9
IELT [70] (TMM'23)	26.8	5.6	0.75	7.1
PMRC [71] (CVPR'23)	30.2	7.1	0.85	5.3
Mix-ViT [35] (PR'23)	24.3	5.3	0.65	6.8
SEB [72] (TPAMI'23)	32.7	7.8	0.95	7.5
EFM-Net [44] (TGRS'23)	28.9	6.3	0.82	5.4
An-Eru [73] (CVPR'23)	23.5	5.1	0.68	3.6
MP-FGVC [74] (AAAI'24)	27.6	6.0	0.78	4.3
FET-FGVC [37] (PR'24)	35.4	8.2	1.05	8.1
ACC-ViT [48] (PR'24)	26.2	5.7	0.72	6.9
MFAL [75] (TGRS'24)	29.8	6.8	0.88	5.8
VMamba [76] (ArXiv'24)	33.1	7.5	0.92	8.2
MetaIRNet [77] (TPAMI'24)	38.6	9.1	1.15	16.4
GCL [78] (TIP'25)	31.5	7.2	0.89	7.1
Ours	31.2	6.2	0.9	7.6

(31.2M) and GFLOPs (6.2) are comparable to the average levels among recent fine-grained classification models, as most methods—including ours—are built upon transformer-based backbones.

Due to the incorporation of image-pair interaction during training, our method incurs a slightly higher training time per epoch (0.9 h), which is above the average. However, this additional cost is limited to the training phase, since the AIPI module is not involved during inference.

The BNTM introduces an additional inference cost by performing hierarchical patch selection, but this increase remains moderate. The BNTM operates in a lightweight manner—each node only involves shallow operations (multihead attention and a few linear layers), and only a single decision path is traversed during inference. Consequently, our method achieves an effective balance between accuracy, interpretability, and computational efficiency, making it well-suited for practical applications.

V. CONCLUSION

In this article, we proposed a novel approach to address the challenges of fine-grained aircraft classification in remote sensing, with a focus on enhancing interpretability and robustness. Our approach integrates three key components: the

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DAFM, which optimally combines attention maps from different layers; the AIPI, which dynamically adapts feature interactions based on intraclass and interclass similarities; and the BNTM, which hierarchically selects informative image patches, creating a clear decision-making path. Experimental results demonstrate that our approach effectively mitigates issues of small interclass variance and large intraclass variance, improving classification accuracy while providing transparency in the decision-making process. This work advances the field of remote sensing image classification by not only boosting performance but also addressing the critical need for model interpretability, making it highly suitable for high-stakes applications. Future research can build upon our framework by exploring further enhancements in adaptive feature interaction and interpretability.

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